Sant Kumar Singh

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Profile

Final-year B.Tech Computer Science (Cyber Security) student with hands-on experience in network security, Active Directory analysis, and log monitoring. Skilled in threat detection, vulnerability assessment, and analyzing attacker behavior using industry-standard tools such as Wireshark, Nmap, and Burp Suite. Passionate about building security automation tools and understanding real-world attack vectors. Interested in SOC operations, incident response, and blue-team cybersecurity roles.

Education

B.Tech in Computer Science (Cyber Security) 2022 – Present
Noida Institute of Engineering and Technology, Greater Noida

Senior Secondary (PCM) 2020 – 2021
Shah Faiz Public School, Uttar Pradesh

Skills

- **Programming:** Python (Scripting, Automation, Tool Development), C++ (Fundamentals, Data Structures)
- **Cyber Security:** Network Security, Threat Detection, Vulnerability Assessment, Active Directory Security, Penetration Testing (Basic)
- **SOC Concepts:** Log Analysis, SIEM, Incident Response, Threat Monitoring, Alert Triage, Event Correlation
- **Tools:** Nmap, Wireshark, Burp Suite, SharpHound, Neo4j, Metasploit (Basic), Splunk (Basic)
- **Networking:** TCP/IP, DNS, HTTP/HTTPS, Firewalls, VPN, OSI Model
- **Operating Systems:** Linux (Ubuntu, Kali), Windows Server, Active Directory Administration

Projects

SOC Log Monitoring and Alert System *SIEM, Python, Log Analysis*

- Collected and analyzed Windows/Linux system logs (authentication, system events) for real-time security monitoring.
- Detected suspicious activities such as multiple failed login attempts, brute-force patterns, and privilege escalations.
- Created alert rules using SIEM concepts to trigger notifications on anomalous behavior with configurable thresholds.
- Simulated attack scenarios (credential stuffing, lateral movement) and analyzed logs to map attacker TTPs.
- Generated structured incident reports with severity classification to support threat investigation workflows.

BlueHound: AD Misconfiguration Detection *Python, Neo4j, SharpHound*

- Developed a tool to detect Active Directory misconfigurations and privilege escalation paths using graph-based analysis.
- Collected AD data via SharpHound and visualized relationship graphs using Neo4j for attack path mapping.
- Identified attack techniques including Kerberoasting, Pass-the-Hash, and unconstrained delegation abuse.
- Prioritized high-risk accounts and generated actionable remediation reports with risk scoring.

Steganography Tool *Python, Cryptography*

- Built a CLI application to securely hide sensitive data within PNG/JPEG images using LSB steganography.
- Integrated AES-256 encryption before embedding to ensure dual-layer confidentiality and data integrity.
- Implemented password-based key derivation (PBKDF2) to protect hidden payloads from unauthorized extraction.

Achievements

- Completed EC-Council certifications in Ethical Hacking and Network Defense with distinction.
- Built hands-on cybersecurity projects focused on real-world attack and defense scenarios using industry tools.
- Participated in Capture the Flag (CTF) competitions, solving challenges in forensics, web exploitation, and cryptography.
- Active member of the college Cyber Security Club, conducting workshops on network security and ethical hacking.

Certifications

- Ethical Hacking Essentials – EHE (EC-Council, 2024)
- Network Defense Essentials – NDE (EC-Council, 2024)
- Linux Commands & Shell Scripting (Coursera – IBM, 2023)
- Introduction to Cybersecurity (Cisco Networking Academy, 2023)